



Team Kappa

Intermediate Maze—Brisbane Boys' College

The Team:

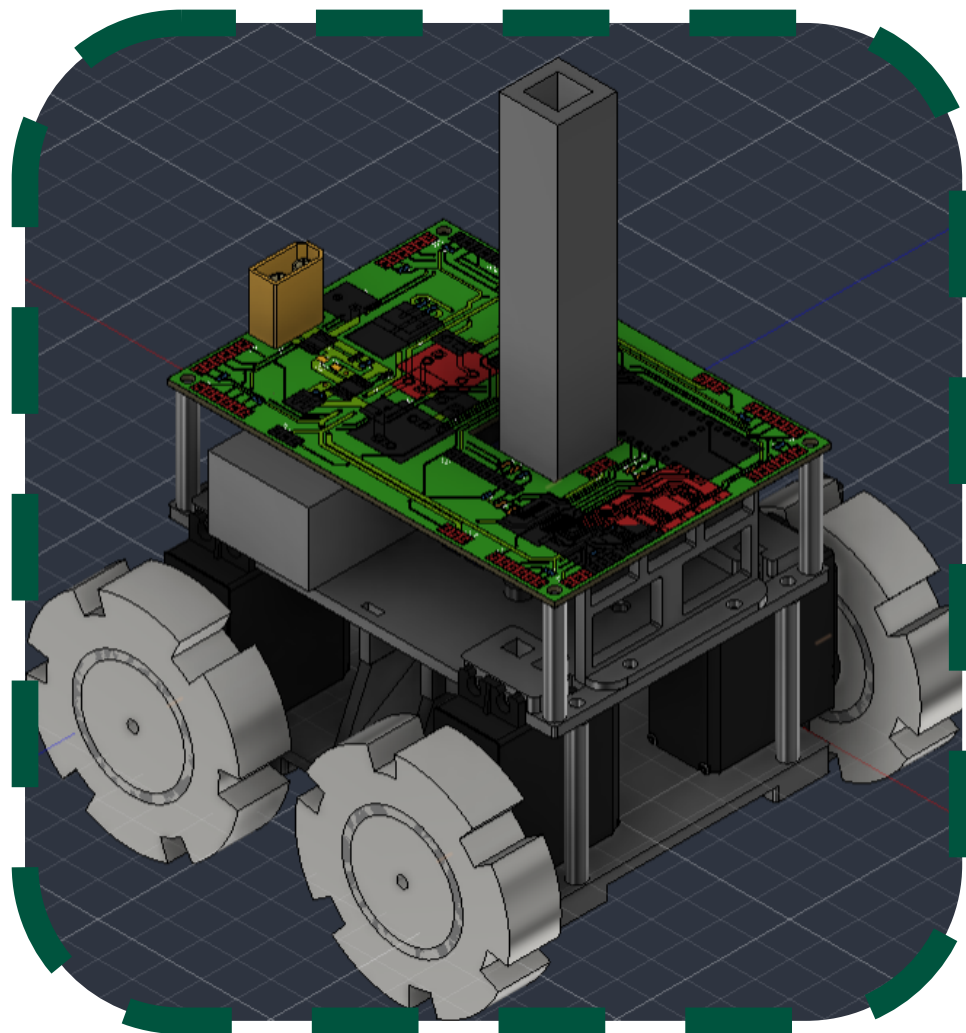
Aryan..... Coder
Chris Design/Solder
Rory Design/Electronics
Joel Coder/Solder

Fun Fact:

This is our first year not doing LEGO

Interesting Feature

It is really small as seen from the image to the right! Measuring 7cm Wide and 11cm



Challenges Endured:

We underestimated the time it takes to learn electronics, C++ code, and make a design fit for printing and manufacturing. However, through determination and focus we managed to bring it together for the competition. Additionally, during our development we made overcomplicated designs and additions that in hindsight were unnecessary. We overcame this through bringing in teammates focused on different areas of the robot to bring as a fresh set of eyes and realise these issues.

Additional Minor Inconveniences We learnt about:

- Solder is hot!
- CAD crashes a lot
- Things seem a lot bigger on screen than in reality
- Datasheets are long

Some of our pseudocode logic:

```
ing state:
- Not black tile & we see opening

1. Turning left state:
- if there is opening on the left
  * Use gyro to turn 90deg left
- Move foward 1 tile
return();

2. Turning right state:
- no opening front, left, there is opening on right:
  * Turn 90deg right
- move foward 1 tile
return();

3. Turning back state:
- no opening front, back, right:
  * turn 180deg
return();

While true():{
  mapping state:

  Check for:
    rescue
    checkpoint
    turning
    moving state
```

Lessons Learnt:

Through making this robot, we learnt to position ourselves in places that interest us and work together well. Another idea we learnt is to question everything we made—why is it this way and not this way—to improve it.